MSDformer: Multiscale Deformable Transformer for Hyperspectral Image Super-Resolution

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***Abstract*— Deep learning–based hyperspectral image super-resolution (SR) methods have achieved remarkable success, which can improve the spatial resolution of hyperspectral images with abundant spectral information. However, most of them utilize 2-D or 3-D convolutions to extract local features while ignoring the rich global spatial-spectral information. In this article, we propose a novel method called the Multiscale Deformable Transformer (MSDformer) for single hyperspectral image SR (SHSR). The proposed method incorporates the strengths of the convolutional neural network (CNN) for local spatial-spectral information and the Transformer structure for global spatial-spectral information. Specifically, a multiscale spectral attention module (MSAM) based on dilated convolution is designed to extract local multiscale spatial-spectral information, which leverages shared module parameters to exploit the intrinsic spatial redundancy and spectral attention mechanism to accentuate the subtle differences between different spectral groups. Then a deformable convolution-based Transformer module (DCTM) is proposed to further extract the global spatial-spectral information from the local multiscale features of the previous stage, which can explore the diverse long-range dependencies among all spectral bands. Extensive experiments on three hyperspectral datasets demonstrate that the proposed method achieves excellent SR performance and outperforms the state-of-the-art methods in terms of quantitative quality and visual results. The code is available at https://github.com/Tomchenshi/MSDformer.git.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral image, super-resolution (SR), Transformer.**

I. Introduction

HYPERSPECTRAL images contain reflectance information of the same scenery in almost continuous tens or even hundreds of narrow spectral bands [1]. They provide more spectral bands with abundant spectral information than standard RGB images as well as multispectral images (MSIs) and can reveal the detailed intrinsic properties and the subtle differences of real objects that are difficult to show in MSIs. Therefore, hyperspectral images are widely used in computer vision and remote sensing, such as classification [2], [3], medical diagnosis [4], detection [5], and geological exploration [6].

However, limited by the amount of incident energy reaching the sensor, there is a trade-off between the spatial resolution and spectral resolution of the captured hyperspectral images [7]. In order to acquire the rich spectral information, the hyperspectral images usually have to sacrifice spatial resolution so that the spatial resolution of hyperspectral images tends to be lower than that of normal RGB images, which may inhibit the image analysis and further practical applications of the hyperspectral images. Due to the difficulty of improving the hardware sensor, the development of algorithms to improve the spatial resolution of hyperspectral images is a very important challenge in the field of image processing.

Super-resolution (SR) aims to improve the spatial resolution of the input image and reconstruct a low-resolution (LR) image with coarse details into a high-resolution (HR) image with fine textures [8]. Therefore, the problem of low spatial resolution of hyperspectral images can be solved by leveraging this technique. According to the number of input images, they can be classified as fusion-based hyperspectral image SR [9], [10] and single hyperspectral image SR (SHSR) [11], [12]. The fusion-based method uses HR auxiliary images such as panchromatic image, RGB image, and MSI to fuse with LR hyperspectral images. Although the fusion-based method is generally popular and often achieves better results than SHSR, they require that the input LR hyperspectral images and HR auxiliary images are in the same scene and rigorously coregistered, which increases the difficulty for the practical application.

Since the SHSR requires no additional auxiliary information in the reconstruction process, it yields good results in most cases and has promising application prospects. The SHSR can be categorized as traditional methods and deep learning–based methods. Most traditional methods such as sparse representation-based methods [13], [14] and low-rank matrix-based methods [15], [16] utilize different prior information as regularization terms to constrain the reconstruction. However, handcrafted methods have been proven to have limited SR performance [17], [18], [19].

Recently, some deep learning–based SHSR algorithms are proposed to achieve superior results, most of which are dedicated to designing modules for the deep network or adding constraints for training loss, for example, 3-D convolution [17], [20], [21], spectral angle map (SAM) loss

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[22], and attention mechanisms [23], [24]. Especially, the grouping-based methods [22], [25], [26] are proposed to explore the spectral correlations of adjacent spectral bands, which exploit the 2-D or 3-D convolutions to extract the local spatial-spectral features. Nevertheless, 2-D convolution can only extract spatial information without considering spectral information. Although 3-D convolution can extract the local spectral information, it ignores the long-range spatial-spectral dependencies that may contribute to the restoration of clear textures and structures and requires much memory, which is more evident at large-scale factors. In addition, most of these methods only extract the simple spatial-spectral feature at the same scale, which wastes the potential multiscale spatial-spectral representations in the multiscale space.

Transformer architecture becomes very popular because of its remarkable performance in visual tasks [27], [28], [29]. Transformer network can capture the internal similarity of the input image. And many Transformer-based methods [30], [31] for low-level vision have demonstrated the superiority and feasibility of Transformer architecture in the field of image restoration. However, little work focuses on the Transformer architecture for SHSR. The main reason is that it is difficult to train an effective Transformer method for SHSR with limited available training samples in the hyperspectral datasets. In addition, traditional self-attention in Transformer explores the long-range relations among spatial dimension while ignoring the spectral similarity. To alleviate the shortcomings of the Transformer on limited datasets, some methods attempt to exploit the complementarity of convolutional neural network (CNN) and Transformer network [30], [32], [33], [34]. Image restoration using swin transformer (SwinIR) [30] and spatial-spectral transformer (SST) [34] put the convolutional layers at the tail of multiple Transformer modules to employ the inductive bias. To further fully exploit the above complementary, Conformer [32] adopted a hybrid parallel network structure to take advantage of convolutional operations and self-attention mechanisms. Based on the above architecture, Interactformer [33] adopts the architecture that combines the 3-D convolution and Transformer block to simultaneously extract the complementary spatial-spectral features. Nevertheless, this work severely requires high hardware memory and computational complexity due to the numerous 3-D convolutions and the parallel architecture.

Taking all the above factors into account, we propose a multiscale deformable Transformer (MSDformer) for SHSR that effectively exploits the complementarity of CNN and Transformer architecture to extract abundant local and global spatial-spectral feature information. Specifically, in terms of hyperspectral image, there is a great deal of spectral correlation and spatial redundancy between different spectral bands. For the adjacent spectral bands, a multiscale spectral attention module (MSAM) is designed to learn the local spatial-spectral interactions. We first divide the LR hyperspectral image into several overlap groups and design a multiscale spatial module based on the dilated convolution (with shared module parameters) to extract the redundant multiscale spatial feature from each spectral group, which can effectively reduce the severe computation and memory consumption [25], [35]. Using the dilated convolution can increase the receptive field without increasing the network parameters. Here, we employ the dilated convolution to extract the multiscale features from different receptive fields. Then, a spectral attention mechanism is introduced to accentuate the subtle discrepancy from the different spectral groups, which is decisive in further improving the SR performance. For the long-range spectral bands, we propose a deformable convolution-based Transformer module (DCTM) to extract the global spatial-spectral information in all spectral bands of the previous stage. Benefiting from the flexible receptive field of the deformable convolution, DCTM can learn variable features for different spectral characteristics, which enhances the global representation capability of the self-attention mechanism. In the proposed deformable convolution-based multihead self-attention (DMSA), the features extracted by 2-D deformable convolution are directly used as query tokens to excavate the diverse spectral differences among all spectral bands. In this way, our method learns not only the local intrinsic spatial-spectral information but also the global diverse spatial-spectral correlations. Compared to other techniques that combine CNN and Transformer, we provide a new architecture of combining CNN and Transformer for SHSR, which can achieve good SR performance with and alleviate the problem of memory and computational consumption in the previous works such as Interactformer [33]. The qualitative and quantitative experiments on three hyperspectral datasets demonstrate the superiority of our method compared with other state-of-the-art methods. In summary, the main contributions of our work are as follows.

1) A novel SHSR method MSDformer is proposed to sufficiently extract the abundant local and global multiscale spatial-spectral information by incorporating the advantages of CNN and Transformer structure, which achieves an excellent SR performance.
2) An MSAM is designed to learn the local multiscale spatial-spectral interactions, which not only exploits the intrinsic spatial redundancy between the spectral groups, but also accentuates the subtle discrepancy of different spectral groups.
3) Considering the inherent correlations of long-range spectra, a DCTM is designed to capture the diverse global spectral dependencies among all spectral bands.

The remaining sections of this article are organized as follows. Section II briefly reviews recent hyperspectral images SR methods. Section III describes the details of the proposed method. Section IV shows the settings and experimental results involving ablation analysis, qualitative experiments, and quantitative experiments. Section V concludes this article.

II. Related Works

In recent years, deep CNNs have achieved remarkable SR performance in nature images. Dong et al. [36] pioneered a three-layer CNN (SRCNN) for natural image SR task, which performed better than traditional methods and attracted great popularity. Subsequently, residual dense network (RDN) [37] combined the dense connection and residual connection to exploit the multilevel features. With the development of the attention mechanism, residual channel attention network (RCAN) [38] introduced the channel attention (CA) mechanism in the SR networks, which markedly improved the

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performance. Then, more attention-based methods such as the second-order attention network (SAN) [39] and the holistic attention network (HAN) [40] were proposed to further enhance the representation capacity of the network. By introducing the nonlocal attention in the network, Mei et al. [41] proposed cross-scale non-local attention module (CSNL) to learn the long-range dependencies among different scales of the images.

Most recently, with the extraordinary success of Transformer in computer vision [27], [42], people have proposed many Transformer architectures for low-level image processing. Liang et al. [30] proposed SwinIR for natural image restoration by adopting Swin Transformer, which achieved incredible results. Other Transformer-based methods such as pre-trained image processing Transformer (IPT) [43] and Restormer [31] were also proposed to exploit the self-attention to improve the restoration performance.

Most of the above methods cannot be directly applied to hyperspectral image SR. They can only utilize the spatial information and little spectral information in a band-by-band manner, which may result in spectral distortion. And they will require excessive computation in a whole-band manner due to the massive spectral bands.

SHSR has attracted widespread attention due to its better availability in practice. Several SHSR methods based on deep learning were successively proposed due to their excellent achievements. Yuan et al. [44] exploited transfer learning and collaborative nonnegative matrix factorization (CNMF) to enforce the collaborations between LR HSIs and HR HSIs. Xie et al. [45] applied the CNMF in the deep neural network to extract the spectral characteristic for real-scene SHSR. Although the aforementioned multistep methods achieved good results, they treated the spatial and spectral information separately and required the handcrafted prior, which were time consuming and unstable. To address these problems, Mei et al. [17] proposed the 3-D fully CNN (3DFCNN) to exploit both the spatial and spectral information, which needed large memory consumption and computational complexity. Subsequently, Li et al. [22] explored the correlation among spectral groups and proposed a grouped deep recursive residual network (GDRRN) by the global residual connection. Inspired by the GDRRN, Jiang et al. [25] proposed a spatial-spectral prior SR (SSPSR) method by the grouping strategy, which achieved a landmark result in a progressive way. Li et al. [20] proposed a mixed convolution network (MCNet) to extract the spatial-spectral features by 2-D and 3-D convolution, which resulted in heavy computational costs when using 3-D convolution for all bands. To further extract the informative spatial features and avoid spectral disorder, Liu et al. [24] proposed a spectral grouping and attention-driven RDN (SGARDN) by introducing the spectral grouping scheme and spectral attention into the network. Wang et al. [26] proposed a group-based recurrent feedback SR network (RFSR) to study the correlation of bands and maintain the spatial-spectral structure by the consistency regularization module. Wang et al. [35] designed a group-based embedding learning and integration network (GELIN) to alleviate the difficulty of feature extraction and reconstruction. Although these methods performed well in SHSR and demonstrated the effectiveness of grouping strategy, they ignored the global long-range spectral correlations and rich multiscale information [46], which were conducive to the restoration of hyperspectral image details.

Transformer has shown the excellent capability to capture the global long-range dependencies in image restoration, but it is difficult to apply the Transformer architecture to hyperspectral image SR due to the mentioned reason in the introduction. Liu et al. [33] attempted to combine the CNN branch and Transformer branch in the network (Interactformer) to simultaneously extract the complementary spatial-spectral features. However, the parallel architecture and 3-D convolution in Interactformer resulted in severe memory consumption and high computational complexity.

To address the above challenge, we propose a novel MSDformer for SHSR that effectively exploits the advantages of the CNN and the Transformer structure to extract abundant local and global multiscale spatial-spectral information.

III. Proposed Method

In this section, we will elaborate the proposed SHSR method MSDformer. It is presented in four parts: overall framework, MSAM, DCTM, and loss function.

A. Overall Framework

There is abundant spatial and spectral information in hyperspectral images. Especially for spectral properties, strong local correlations and long-range dependence are beneficial for detail restoration in SR. Therefore, we propose an MSDformer to simultaneously leverage these relations. As shown in Fig. 1, the proposed MSDformer consists of two parts: CNN structure and Transformer architecture, exploiting MSAM and DCTM to learn the local and global diverse spatial-spectral interactions.

We denote the input LR hyperspectral image as $I_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times B}$ with B spectral bands and spatial resolution of $H \times W$, the original HR hyperspectral image as $I_{\mathrm{HR}} \in \mathbb{R}^{sH\times sW\times B}$, and the output SR hyperspectral image as $I_{\mathrm{SR}} \in \mathbb{R}^{sH\times sW\times B}$, in which s represents the SR scale factor. Our goal is to predict the SR result I_{SR} only from LR input I_{LR} by the proposed network MSDformer and make I_{SR} and I_{HR} as identical as possible. It can be formulated as

$$I_{\mathrm{SR}} = H_{\mathrm{MSDformer}}(I_{\mathrm{LR}}) \tag{1}$$

where $H_{\mathrm{MSDformer}}(\cdot)$ denotes the function of the proposed MSDformer method.

To exploit the strong correlation between the adjacent bands and improve efficiency, many grouping strategy-based methods have been proposed and shown the superiority over other methods. Following these methods, we first divide the input LR hyperspectral image into G groups with overlapping way, $I_{\mathrm{LR}} = [I^1_{\mathrm{LR}}, \ldots, I^g_{\mathrm{LR}}, \ldots, I^G_{\mathrm{LR}}]$. Especially, following the settings in [25] and [35], each hyperspectral image group has c spectral bands with o overlaps. The detailed settings are discussed in Section IV. For the local information, we design an MSAM to extract the multiscale spatial-spectral feature from each group $I^g_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times B/G}$, which efficiently exploits the spatial redundancy and spectral discrepancy between adjacent

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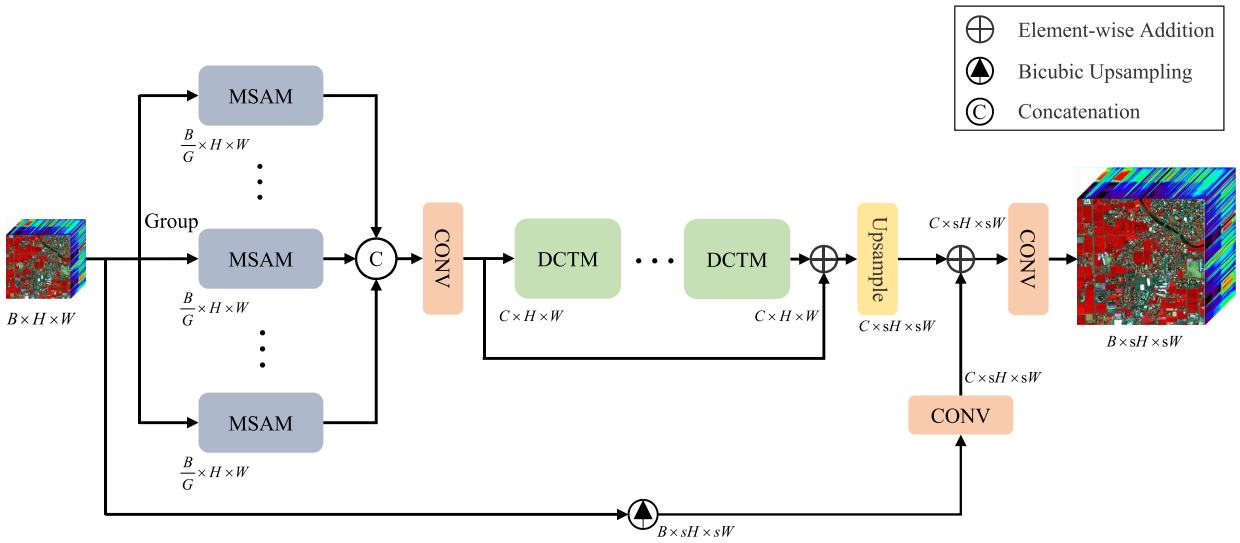


Fig. 1. Overall architecture of the proposed MSDformer consisting of the MSAM, DCTM, and upsampling module.

spectral bands. The obtained multiscale spatial-spectral feature $F_{\text{spc}}^{g} \in \mathbb{R}^{H \times W \times B/G}$ from the gth group can be formulated as

$$F_{\text{spc}}^{g} = H_{\text{MSAM}}\left(I_{\text{LR}}^{g}\right) \tag{2}$$

where $H_{\text{MSAM}}(\cdot)$ denotes the function of MSAM. Then all the above group features are concatenated together to obtain the local multiscale spatial-spectral feature $F_{\text{local}} \in \mathbb{R}^{H \times W \times B}$, which can be formulated as

$$F_{\text{local}} = \text{Cat}\left(F_{\text{spc}}^{1}, \ldots, F_{\text{spc}}^{g}, \ldots, F_{\text{spc}}^{G}\right) \tag{3}$$

where $\text{Cat}(\cdot)$ denotes the concatenating function in the spectral dimension.

To fully utilize the global information, we first map the obtained local feature F_{local} to a high spectral dimension, which can be formulated as

$$F_0 = H_0(F_{\text{local}}) \tag{4}$$

where $F_0 \in \mathbb{R}^{H \times W \times C}$ denotes the spatial-spectral features of high spectral dimension with C spectral bands and spatial resolution of $H \times W$ and $H_0(\cdot)$ denotes the 1×1 convolution function. C is demonstrated in the implementation details.

Besides the above local correlation between adjacent spectral bands, the global spatial-spectral interrelationship is also essential for SR performance. So we design a DCTM to capture the diverse long-range spectral dependencies from the feature F_0 of high spectral dimensionality. The global deep spatial-spectral feature F_N from the Transformer architecture is extracted by N DCTMs, which can be denoted as

$$\begin{aligned} F_N &= H_{\text{DCTM},N}(F_{N-1}) \\ &= H_{\text{DCTM},N}\left(H_{\text{DCTM},N-1}\left(, \ldots, \left(H_{\text{DCTM},1}(F_0), \ldots, \right)\right)\right) \end{aligned} \tag{5}$$

where $H_{\text{DCTM},N}$ and $F_N \in \mathbb{R}^{H \times W \times C}$ denote the Nth DCTM and the global deep spatial-spectral feature, respectively. Then the local residual learning is introduced to get the fused deep spatial-spectral feature $F_f \in \mathbb{R}^{H \times W \times C}$, which can be denoted as

$$F_f = F_N + F_0. \tag{6}$$

The upsampling module is used to increase the spatial size of the fused deep spatial-spectral features F_f, which can be denoted as

$$F_{\text{up}} = H_{\text{up}}\left(F_f\right) \tag{7}$$

where $H_{\text{up}}(\cdot)$ denotes the upsampling function that is conducted by the PixelShuffle [47] followed by a convolutional layer and F_{up} with C spectral bands and spatial size of $sH \times sw$ is upscaled features.

Finally, to reduce the difficulty of training the network and focus on the high-frequency information, the upsampled features via the bicubic function are directly added to the tail of the network. The 1×1 convolution following the bicubic is used to adjust the spatial size for the addition operation. And the 1×1 convolution following the addition is used to make the spatial size consistent with the original HR hyperspectral image. The above upsampling process can be denoted as

$$F_{\text{res}} = F_{\text{up}} + H_1(I_{\text{LR}} \uparrow) \tag{8}$$

$$I_{\text{SR}} = H_2(F_{\text{res}}) \tag{9}$$

where $F_{\text{res}} \in \mathbb{R}^{sH \times sW \times C}$ is the final upsampled feature, $H_1(\cdot)$, and $H_2(\cdot)$, denote 1×1 convolutional layer, $I_{\text{LR}} \uparrow$ is the bicubic upsampling version of the input LR hyperspectral image, and I_{SR} with B spectral bands and spatial size of $sH \times sW$ is the reconstructed HR hyperspectral image.

B. Multiscale Spectral Attention Module

Some natural image SR methods only consider the spatial information, which is not suitable for hyperspectral images. In hyperspectral images, it is necessary to take into account not only the spatial information but also the rich spectral information. Especially for the spectral groups, there are massive spatial similarities and slight spectral differences between

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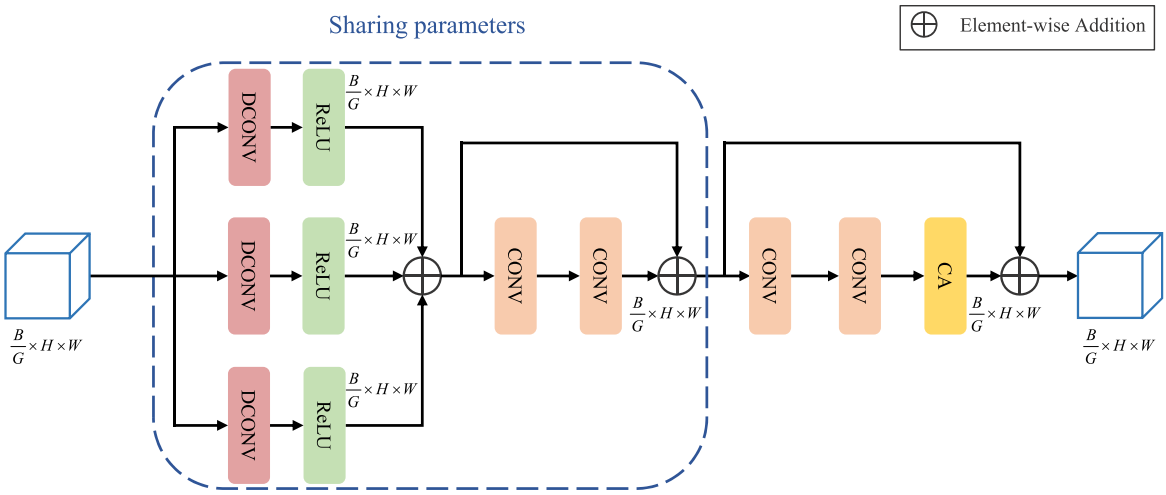


Fig. 2. Structure of the designed MSAM.

different spectral groups. And most SHSR methods adopt the 2-D or 3-D convolutions to extract the spatial-spectral information in single-scale space, which ignores the abundant multiscale mapping relationships in multiscale space.

Therefore, as shown in Fig. 2, we design an MSAM to extract the local multiscale spatial-spectral features, considering the plenty of spatial similarities and subtle spectral discrepancy in spectral groups. Dilated convolution is designed to increase the receptive field without increasing the network parameters, which is used to extract the multiscale features from different receptive fields in MSAM. In addition, a larger receptive field enables the network to effectively find similar image patches for recovering clear textures. So we first adopt three dilated convolutions with different dilation rates (i.e., 1, 3, and 5, respectively) followed by rectified linear unit (ReLU) activation function to extract the multiscale spatial information and explore the spatial similarity between the multiscale features, which facilitates the recovery of the details. Let $I_{\mathrm{LR}}^{g} \in \mathbb{R}^{H\times W\times B/G}$ denote the input of the gth MSAM, and the extracted features F_i^g at different scales can be formulated as

$$\begin{aligned} F_1^g &= H_D\left(I_{\mathrm{LR}}^g, 1\right) \\ F_2^g &= H_D\left(I_{\mathrm{LR}}^g, 3\right) \\ F_3^g &= H_D\left(I_{\mathrm{LR}}^g, 5\right) \end{aligned} \tag{10}$$

where $H_D(\cdot,\cdot)$ denotes the function of dilated convolution. Then the obtained features at different scales are fused and further extracted by the residual spatial module, which can be denoted as

$$\begin{aligned} F_d^g &= F_1^g + F_2^g + F_3^g \\ F_{\mathrm{spa}}^g &= F_d^g + H_3\left(F_S^g\right) \end{aligned} \tag{11}$$

where $F_d^g \in \mathbb{R}^{H\times W\times B/G}$ denotes the fused multiscale feature, $H_3(\cdot)$ denotes the function of two 3×3 convolutional layers, and F_{spa}^g denotes the residual multiscale feature. The combination of abundant spatial-spectral features at different scales can explore more comprehensive receptive fields as much as possible, which is beneficial for extracting more details. Considering the spatial redundancy, we adopt the sharing parameter strategy on each group to extract the above multiscale spatial information. In addition, to exploit the similarity and complementarity in spectral dimension, the residual CA block (RCAB) [38] is adopted to extract the different spectral characteristics by excavating the interactions across feature spectra, which can be denoted as

$$F_{\mathrm{spc}}^g = H_{\mathrm{RCAB}}\left(F_{\mathrm{spa}}^g\right) \tag{12}$$

where $F_{\mathrm{spc}}^g \in \mathbb{R}^{H\times W\times B/G}$ denotes the residual multiscale spatial-spectral feature. The RCAB is proposed to enhance the SR performance for natural image and has been widely used as the basic block in hyperspectral image SR for the extraction of spectral information. Specifically, the RCAB consists of two convolutional layers and a middle ReLU layer, followed by a CA module. The CA module consists of a global pooling operation, a dimensionality-reduction layer, a dimensionality-increasing layer, and a sigmoid activation function, which can explore the cross-channel correlations. And the reduction ratio is adopted to control the number of feature channels. Finally, the multiplication between the weight on channel dimension and the multiscale spatial feature F_{spa}^g is exploited to adaptively calibrate the importance of each channel.

Through the MSAM, we extract the local multiscale spatial-spectral information by studying the interactions between local adjacent spectral groups, which supports the later extraction of global information.

C. Deformable Convolution-Based Transformer Module

Some methods based on the self-attention and Transformer structure for natural images have achieved excellent performance, which has demonstrated the effectiveness of long-range spatial dependence in SR. The success of Interactformer proves that both the local spatial-spectral information in CNN structure and the long-range spatial-spectral information in Transformer structure are very important for hyperspectral image SR. So we design a DCTM to learn the diverse long-range correlations among all spectral bands.

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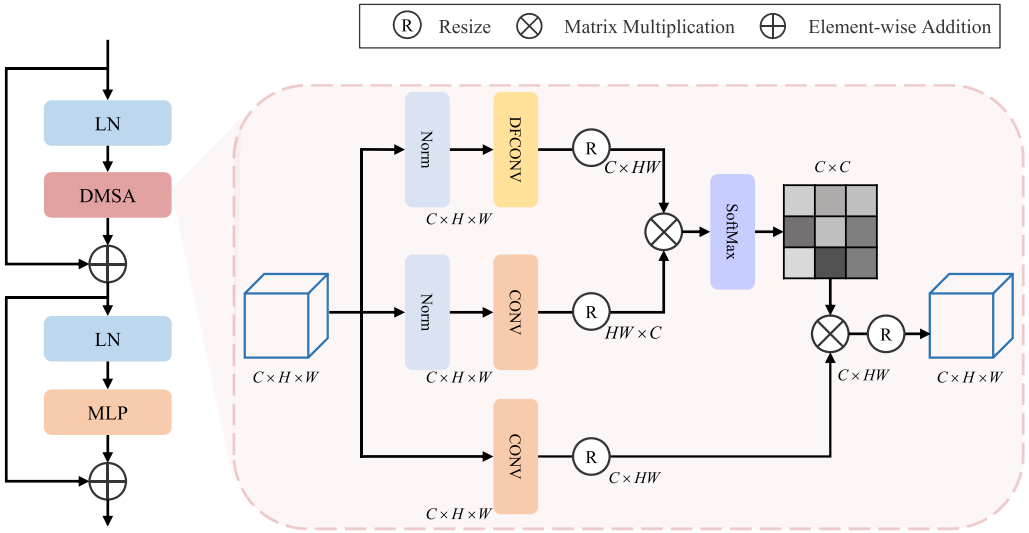


Fig. 3. Structure of the designed DCTM.

As shown in Fig. 1, we first map the extracted local spatial-spectral features from the previous CNN structure to a high-dimensional spectral space with C bands (i.e., F_0). Then N successive DCTMs are used to extract the global spatial-spectral features from the F_0. The DCTM consists of the DMSA and a 3×3 convolution, which exploit the flexible global spectral modeling capabilities and the characteristic of inductive bias in the convolutional layer. SwinIR [30] has shown the effectiveness of the structure of Transformer block followed by convolution, which can enable the network to keep the advantages of CNN structure and Transformer architecture. It allows the Transformer-based network to be well trained and learn good mapping with limited data, which is ideal for hyperspectral image SR. In addition, the convolutional layer is not adopted in the first DCTM. As shown in Fig. 3, following most of the Transformer encoder modules, DCTM consists of two residual structures, one of which is the LayerNorm layer followed by the deformable convolution-based self-attention layer, and the other is the LayerNorm layer followed by the multilayer perception (MLP) layer. The input of the nth DCTM is $F_n \in \mathbb{R}^{H\times W\times C}$ and the process of DCTM can be denoted as

$$\begin{aligned} F^g_{\text{global}} &= \text{DMSA}(\text{LN}(F_n)) + F_n \\ F_{n+1} &= \text{MLP}\left(\text{LN}\left(F^g_{\text{global}}\right)\right) + F^g_{\text{global}} \end{aligned} \tag{13}$$

where $F^g_{\text{global}} \in \mathbb{R}^{H\times W\times C}$ denotes the middle global feature of DMSA, $\text{LN}(\cdot)$ denotes the function of LayerNorm layer, and $\text{MLP}(\cdot)$ denotes the function of MLP layer.

Considering the various spectral correlations and differences among all spectral bands, we design an adaptive multihead self-attention (MHSA) scheme based on the deformable convolution to learn the specific weights for different spectral characteristics. Specifically, we use the deformable convolution [48] as the query in the self-attention, which can reveal the diverse global spectral interaction. Let X denote the input of the deformable convolution, and the value of deformed feature X' at position (i, j) can be formulated as

$$X'_{(i,j)} = \sum_{c=0}^{C-1} w^c_{(i,j)} \cdot X\left(p^c_{(i,j)} + p^c_n + \Delta p_{(i,j)}\right) \tag{14}$$

where C is the channel numbers of X', $w^c_{(i,j)}$ denotes the weight of deformable convolution on the cth channel, $p^c_{(i,j)}$ denotes the spatial coordinate (i, j) of the target sampling position on the cth channel of the input feature X', p^c_n denotes the set of all sampling positions on the cth channel of the standard $k \times k$ convolution kernel, and $\Delta p_{(i,j)}$ denotes the learned offset. Since $[p^c_{(i,j)} + p^c_n + \Delta p_{(i,j)}]$ may be fractional, the bilinear interpolation strategy is introduced like [49]. Compared with the fixed receptive fields of standard convolution, deformable convolution provides more flexible and effective receptive fields for the target task, which facilitates the assembly of more important information.

In a DMSA module, we denote $F_n \in \mathbb{R}^{H\times W\times C}$ as the input of DMSA, and F_n is first projected and reshaped into the query ($Q \in \mathbb{R}^{C\times HW}$), key ($K \in \mathbb{R}^{C\times HW}$), and value ($V \in \mathbb{R}^{C\times HW}$) by the deformable convolution and two 1×1 pointwise convolutions, respectively. It can be formulated as

$$Q = W_{\text{deformable}}(F_n), \quad K = W_k(F_n), \quad V = W_v(F_n) \tag{15}$$

where $W_{\text{deformable}}$ denotes the function of deformable convolution, and W_k and W_v denote the function of pointwise convolutions. It is worth noting that, unlike the general spatialwise self-attention mechanism, the DMSA calculates the self-attention matrix ($M \in \mathbb{R}^{C\times C}$) in the spectral dimension. Similar to the standard MHSA, we split the Q, K, V into D heads to represent as many different mappings as possible. For the sake of simplicity, we only show the calculation process of single head, which can be denoted as

$$\text{DMSA}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{16}$$

where DMSA is the output of single head, and d_k is the dimension of Q, specifically as C/N. Finally, the outputs of all heads are concatenated in spectral dimension.

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Through the above channel-aware self-attention algorithm in DCTM, we can accurately exploit the various similarities and complementarity between different spectral bands for the recovery of details in SHSR.

D. Loss Function

Substantial works have demonstrated the positive effect of l_1 and l_2 losses in SR tasks [50]. Since the l_2 loss often results in excessively smooth results where l_1 loss provides a more balanced error distribution and better convergence, we adopt the $l1$ loss to measure the accuracy of the reconstructed SR image. The l_1 loss calculates the pixelwise difference between the reconstructed SR hyperspectral images and original HR hyperspectral images, which can be formulated as

$$\mathcal{L}_1(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\left\|H_{\text{hr}}^n - H_{\text{sr}}^n\right\|_1 \tag{17}$$

where N is the number of the images in a training batch, Θ denotes the parameter set of our proposed network, and H_{hr}^n and H_{sr}^n denote the nth reconstructed SR hyperspectral image and original HR hyperspectral image, respectively. Although the l_1 loss can achieve good performance for natural image SR, it ignores the spectral characteristic in hyperspectral image and may result in spectral distortion. We introduce the SAM loss to preserve the spectral consistency and spatial details. The SAM loss can be formulated as

$$\mathcal{L}_{\text{spe}}(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{\pi}\arccos\left(\frac{H_{\text{hr}}^n \cdot H_{\text{sr}}^n}{\left\|H_{\text{hr}}^n\right\|_2 \cdot \left\|H_{\text{sr}}^n\right\|_2}\right). \tag{18}$$

In addition, inspired by Wang et al. [26], we introduce the gradient information to further enhance the sharpness of the reconstructed image, which is calculated by the difference between adjacent pixels. The gradient map of a hyperspectral image H is formulated as

$$\begin{aligned}\nabla H &= (\nabla_h H, \nabla_w H, \nabla_l H)\\ M(H) &= \|\nabla H\|_2\end{aligned} \tag{19}$$

where $M(\cdot)$ is the operator to obtain the gradient map of H, and ∇_h, ∇_w, and ∇_l denote the function to compute the gradient value along horizontal, vertical, and spectral dimensions. We leverage the gradient loss to decrease the distance between the gradient map extracted from the reconstructed SR image and the original HR image, which facilitates the local sharpness of the upscaled images. The gradient loss can be formulated as

$$\mathcal{L}_{\text{gra}}(\Theta) = \frac{1}{N}\sum_{n=1}^{N}\left\| M\left(H_{\text{hr}}^n\right) - M\left(H_{\text{sr}}^n\right)\right\|_1. \tag{20}$$

Finally, the total loss for training the proposed network can be denoted as

$$\mathcal{L}_{\text{total}}(\Theta) = \mathcal{L}_1 + \lambda_1\mathcal{L}_{\text{spe}} + \lambda_2\mathcal{L}_{\text{gra}} \tag{21}$$

where λ_1 and λ_2 denote the tradeoff parameters of different losses. Specifically, λ_1 controls the weight of the spectral term to avoid the too large value, which results in excessive focus on spectral similarity at the expense of spatial detail. Similarly, λ_2 controls the weight of the gradient term to avoid excessive sharpening in both spatial and spectral structures. Following [26] and [35], the combination of the above losses can achieve excellent SR performance in both spatial and spectral dimensions. λ_1 is set to 0.5 and λ_2 is set to 0.1, empirically.

IV. Experiments and Results

A. Implementation Details

In the proposed network MSDformer, the input hyperspectral images are split into G groups with o overlaps between adjacent groups. Each group has c spectral bands, and we select the last c bands for the last group. The kernel size of convolution for extracting the spatial feature is set to 3×3, and the size of others for the expansion and reduction of the channel is set to 1×1 like in CA. Zero padding is adopted to maintain the spatial size of feature maps. Following the settings in [38], we set the reduction ratio in CA as 16. In MSAM, the dilation rate of three dilated convolutions is set to 1, 3, and 5, respectively. In DCTM, the number of feature maps C is set to 240, and the number of successive DCTMs is set to 4, which is discussed in the ablation study. The progressive upsampling strategy by PixelShuffle [47] is adopted to magnify the spatial size of input LR hyperspectral images for reducing the parameters (e.g., upsampling three times for scale factor $\times 8$). In the loss function, we set $\lambda_1 = 0.5$ and $\lambda_2 = 0.1$. For the training settings, we use the Adam optimizer with the default setting to train the network for 300 epochs, and the size of mini-batch is set to 32. The initial learning rate is set to $5e^{-5}$ and decays by ten times after 160 epochs. The proposed model is implemented by Pytorch on NVIDIA RTX 3090 GPU.

B. Datasets and Experimental Setup

We evaluate the proposed method on three common real-scenario hyperspectral image datasets, that is, Chikusei [51], Pavia Centre [52], and Houston.[1] The proposed method is compared with five state-of-the-art methods for SHSR, including bicubic, 3DFCNN [17], GDRRN [22], SSPSR [25], RFSR [26]. For the above methods, we try our best to adjust the hyperparameters of them to achieve their best performance on the above datasets at the scale factors $\times 2$, $\times 4$, and $\times 8$.

Six widely used evaluation metrics are utilized to comprehensively demonstrate the SR performance, including peak signal-to-noise ratio (PSNR), structure similarity (SSIM), SAM, cross correlation (CC), root-mean-squared error (RMSE), and erreur relative globale adimensionnelle de synthese (ERGAS). The best values for these indices are $+\infty$ 1, 0, 1, 0, and 0, respectively. The PSNR, SSIM, and RMSE are commonly used in the evaluation of natural image restoration quality that are calculated on the single-channel image, we compute their average values on all spectral bands in hyperspectral images for comparison. The remaining three metrics CC, SAM, and ERGAS are commonly used evaluation metrics in the hyperspectral image fusion task.

[1] https://hyperspectral.ee.uh.edu/?page_id=1075

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TABLE I
QUANTITATIVE PERFORMANCE OVER FOUR TEST IMAGES ON THE CHIKUSEI DATASET AT DIFFERENT SCALE FACTORS. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE REPRESENTS THE SECOND BEST

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	ERGAS↓
bicubic	×2	43.2125	0.9721	1.7880	0.9781	0.0082	3.5981
3DFCNN [17]	×2	45.4477	0.9828	1.5550	0.9854	0.0064	2.9235
GDRRN [22]	×2	46.4286	0.9869	1.3911	0.9885	0.0056	2.6049
SSPSR [25]	×2	47.4073	0.9893	1.2035	0.9906	0.0051	2.3177
RFSR [26]	×2	47.5563	0.9897	1.1508	0.9911	0.0049	2.3130
Ours	×2	**47.6881**	**0.9899**	**1.1457**	**0.9911**	**0.0049**	**2.2388**
bicubic	×4	37.6377	0.8954	3.4040	0.9212	0.0156	6.7564
3DFCNN [17]	×4	38.1221	0.9079	3.3927	0.9276	0.0147	6.4453
GDRRN [22]	×4	39.0864	0.9265	3.0536	0.9421	0.0130	5.7972
SSPSR [25]	×4	39.9797	0.9393	2.4864	0.9528	0.0119	5.1905
RFSR [26]	×4	39.8950	0.9382	2.4656	0.9517	0.0120	5.2334
Ours	×4	**40.0902**	**0.9405**	**2.3981**	**0.9539**	**0.0118**	**5.0818**
bicubic	×8	34.5049	0.8069	5.0436	0.8314	0.0224	9.6975
3DFCNN [17]	×8	34.7274	0.8142	4.9514	0.8379	0.0218	9.4706
GDRRN [22]	×8	34.7395	0.8199	5.0967	0.8381	0.0213	9.6464
SSPSR [25]	×8	35.1643	0.8299	4.6911	0.8560	0.0206	9.0504
RFSR [26]	×8	35.5049	0.8405	4.2785	0.8661	0.0199	8.6338
Ours	×8	**35.5914**	**0.8452**	**4.1381**	**0.8693**	**0.0197**	**8.5203**

C. Experimental Results on Chikusei Dataset

The Chikusei dataset is taken by the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Ibaraki, Japan. It contains 2517 × 2335 pixels with 128 bands in the spectral range from 363 to 1018 nm and the ground sampling distance is 2.5 m. Due to the invalid information in the edge region, following the previous work [25], [26], we crop the informative central region with 2304 × 2048 × 128 of the original scene.

Specifically, we crop the top region of the image into four nonoverlap hyperspectral images with the size of 512 × 512 × 128 for testing. The rest region of the image is cropped to form the overlap patches for training (10% of the training data is randomly selected as a validation set). When the scale factor s is ×2, we extract the patches with the size of 64 × 64 × 128 by the overlapping of 32 pixels. When the scale factor s is ×4, we extract the patches with the size of 128× 128 × 128 by the overlapping of 64 pixels. When the scale factor s is ×8, we extract the patches with the size of 256 × 256 × 128 by the overlapping of 128 pixels. These patches are used as reference HR hyperspectral images and the LR hyperspectral images are generated by bicubic downsampling from them for the corresponding scale factor. SSPSR [25] has studied the influence of the number of spectral bands in each group c and the overlaps o by a large number of experiments. They conclude that too large and too small c will result in bad SR performance while appropriate c could enhance the SR performance. In addition, the proper o is also beneficial to the performance of the network while a large o will increase the computational burden. Therefore, on the basis of the above research, we define the c and o according to the datasets. For the Chikusei dataset, the number of bands c in each group is 8 and the overlap is 2 (i.e., $c = 8$ and $o = 2$).

As shown in Table I, we compare the proposed method with some state-of-the-art methods on Chikusei testing set. The averages of six objective quantitative indices are used to measure the performance of these methods at scale factors ×2, ×4, and ×8. Although 3DFCNN [17] employs the 3-D convolution to extract the spectral and spatial information to some extent, it cannot effectively extract spatial-spectral features and result in large computational complexity. To effectively explore the spectral information, GDRRN [22] adopts the grouping strategy to design a recursive network, which reduces the spectral distortion. Inspired by this scheme, SSPSR [25] and RFSR [26] adopt the designed grouping strategy and achieve the comparable performance both in spatial and spectral similarity. It is worth noting that the proposed MSDformer achieves better results than the other methods for all metrics at scale factors ×2, ×4, and ×8. Specifically, by considering the local and global spectral relations, our method can not only achieve good spatial performance, but also achieve excellent spectral reconstruction performance in terms of SAM.

As shown in Fig. 4, we also visualize the qualitative results of the reconstructed hyperspectral image from Chikusei test set at the scale factor ×4. Specifically, we select the 31st, 98th, and 61st bands of the hyperspectral image in the above image and treat them as the RGB channels for better visualization. It can be noted that early methods such as 3DFCNN and GDRRN [22] tend to produce very blurred edges and details in the reconstructed image. Recent grouping-based methods such as SSPSR [25] and RFSR [26] can produce comparable visual reconstruction results, but there are still some shortcomings for restoring some detailed lines. Our proposed MSDformer restores the better low-frequency and high-frequency details than other methods. The advantages of our method can be better observed based on the enlarged view of the details in the red box. In addition, we further visualize the error images of the above results to verify the advantages of our method in Fig. 5, which intuitively displays the accuracy of the SR reconstruction for each pixel on the test image. The bluer the color of each pixel in error images, the higher the accuracy of the SR reconstruction. Early methods such as 3DFCNN [17] and GDRRN [22] produce more red pixel regions compared to other methods. Although later methods have fewer red pixels than them, our proposed MSDformer provides the least number of red pixels and reconstruction errors in all algorithms (please refer to the area marked with red boxes). Finally, for the spectral aspect, we show the mean spectral difference curve of all test hyperspectral images on Chikusei dataset at the scale factors ×4 and ×8. In Fig. 6, compared with the reflectance of random pixels in the reconstructed image, the mean spectral difference can comprehensively present the accuracy of spectral reconstruction quality among whole image. It shows that our proposed method can achieve the best spectral performance while other methods provide more spectral distortion.

D. Experimental Results on Houston Dataset

The Houston 2018 dataset is the part of the 2018 IEEE GRSS Data Fusion Contest, taken by the ITRES CASI 1500 spectral imager over the University of Houston campus and its surrounding urban area in Houston, TX, USA. It contains 4172 × 1202 pixels with 48 bands in the spectral range

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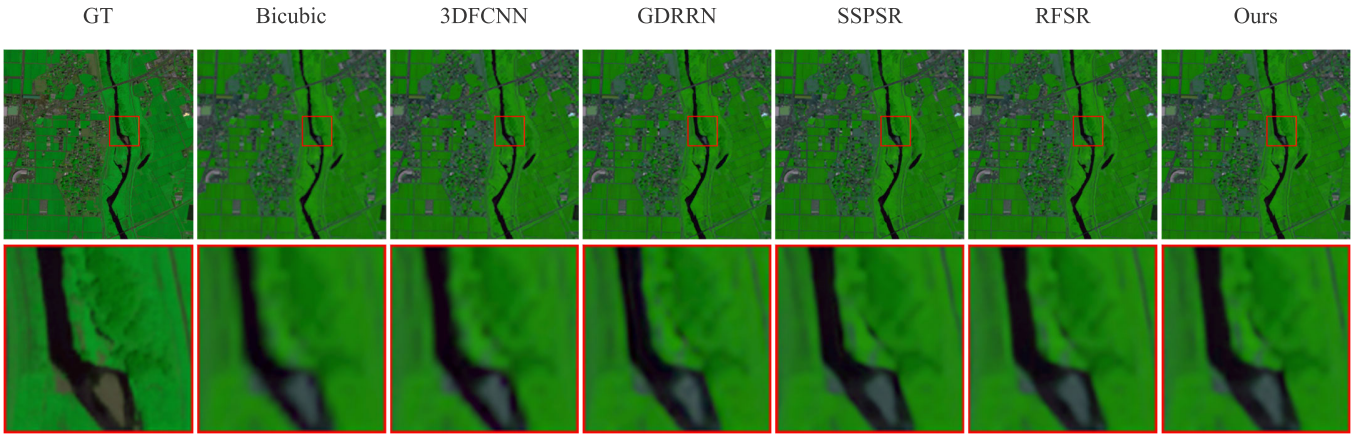


Fig. 4. Reconstructed test hyperspectral image in the Chikusei dataset with spectral bands 31-98-61 as R-G-B at scale factor ×4. (From Left to Right) Ground truth, results of bicubic, 3DFCNN [17], GDRRN [22], SSPSR [25], RFSR [26], and the proposed MSDformer method.

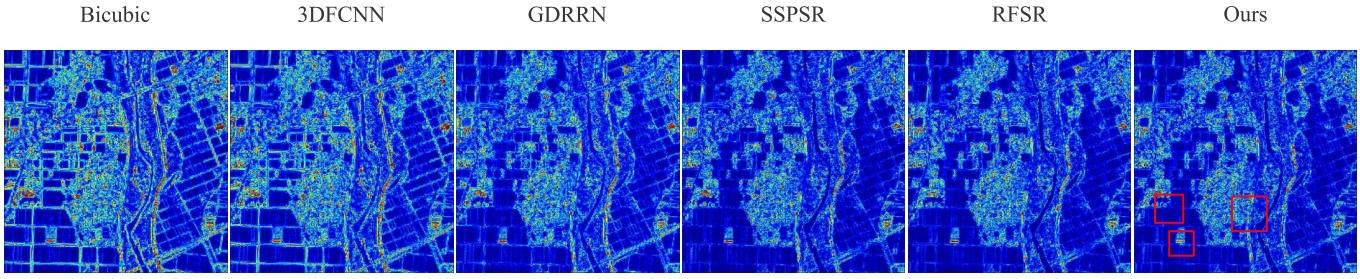


Fig. 5. Error maps of the test hyperspectral image in the Chikusei dataset at the scale factor ×4.

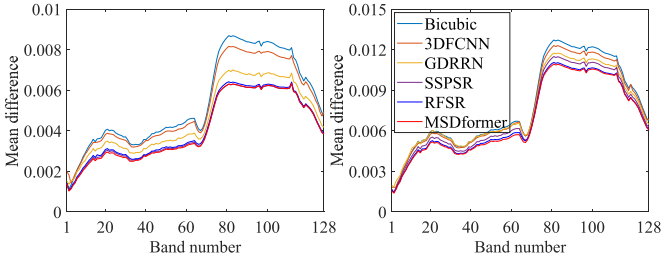


Fig. 6. Mean spectral difference curve of four test hyperspectral images in the Chikusei dataset at the scale factors (Left) ×4 and (Right) ×8, respectively.

from 380 to 1050 nm and the ground sampling distance is 1 m. Similar to the setting of Chikusei, we crop the top region of the image into eight nonoverlap hyperspectral images with the size of 256 × 256 × 48 for testing. The rest region of the image is cropped to form the overlap patches for training (10% of the training data is randomly selected as a validation set). The LR hyperspectral images are generated by bicubic downsampling from these patches for the different scale factors. For the Houston dataset, the number of bands c in each group is 8 and the overlap is 2 (i.e., $c = 8$ and $o = 2$).

In Table II, we show the quantitative results of all methods in terms of six evaluation metrics on the Houston test set at scale factors ×2, ×4, and ×8. The proposed MSDformer achieves the better results compared with the other state-of-art methods for six evaluation metrics at all scale factors. This demonstrates the effectiveness of local and global spatial-spectral information for the hyperspectral images SR.

TABLE II

QUANTITATIVE PERFORMANCE OVER EIGHT TEST IMAGES ON THE HOUSTON DATASET AT DIFFERENT SCALE FACTORS. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE REPRESENTS THE SECOND BEST

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	ERGAS↓
bicubic	×2	49.4735	0.9915	1.2707	0.9940	0.0040	1.3755
3DFCNN [17]	×2	50.7939	0.9941	1.2168	0.9949	0.0034	1.1722
GDRRN [22]	×2	51.5205	0.9949	1.1241	0.9957	0.0031	1.0723
SSPSR [25]	×2	52.1246	0.9954	1.0101	0.9965	0.0029	1.0035
RFSR [26]	×2	52.3828	0.9958	0.9623	0.9967	0.0028	0.9715
Ours	×2	**52.6260**	**0.9960**	**0.9528**	**0.9969**	**0.0027**	**0.9454**
bicubic	×4	43.0272	0.9613	2.5453	0.9741	0.0086	2.9085
3DFCNN [17]	×4	43.2680	0.9669	2.6128	0.9661	0.0079	2.8698
GDRRN [22]	×4	44.2964	0.9730	2.5347	0.9760	0.0069	2.4700
SSPSR [25]	×4	45.6017	0.9778	1.9650	0.9850	0.0063	2.1380
RFSR [26]	×4	45.8677	0.9792	1.8304	0.9858	0.0060	2.0659
our	×4	**46.2015**	**0.9807**	**1.7776**	**0.9870**	**0.0059**	**1.9962**
bicubic	×8	38.1083	0.8987	4.6704	0.9177	0.0152	5.1229
3DFCNN [17]	×8	38.0152	0.9030	4.7085	0.9093	0.0146	5.0865
GDRRN [22]	×8	38.2592	0.9085	4.9045	0.9138	0.0140	4.9135
SSPSR [25]	×8	39.2844	0.9164	4.2673	0.9346	0.0130	4.4212
RFSR [26]	×8	39.4899	0.9211	3.8403	0.9379	0.0126	4.2967
Ours	×8	**39.7452**	**0.9227**	**3.6613**	**0.9417**	**0.0124**	**4.2110**

Specifically, our method can further enhance SR performance both on the spatial and spectral aspects by exploiting the long-range spectral dependencies.

As shown in Fig. 7, we also visualize the qualitative results of the reconstructed hyperspectral image from Houston test

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set at the scale factor ×4. Especially, We treat the 29th, 26th, and 19th bands of the hyperspectral image in the above image as the R-G-B channels for better visualization. Early methods such as 3DFCNN [17] and GDRRN [22] tend to produce very blurred edges and details in the reconstructed image. Recent grouping-based methods such as SSPSR [25] and RFSR [26] can produce comparable visual reconstruction results, but there are still some shortcomings for restoring some detailed lines. Our proposed MSDformer restores clearer low-frequency and high-frequency details than other methods. The advantages of our method can be better observed based on the enlarged view of the details in the red box. Finally, for the spectral aspect, we show the mean spectral difference curve of all test hyperspectral images on Houston dataset at the scale factors ×4 and ×8 in Fig. 8. As the spectral pattern of the Houston dataset is relatively simple compared to the other datasets, our method performs relatively far from other methods in terms of all metrics, which further demonstrates the advantages of our methods in leveraging the spectral similarity and complementarity.

TABLE III

QUANTITATIVE PERFORMANCE OVER FOUR TEST IMAGES ON THE PAVIA DATASET AT DIFFERENT SCALE FACTORS. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE REPRESENTS THE SECOND BEST

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	ERGAS↓
bicubic	×2	32.0583	0.9139	4.5419	0.9491	0.0256	4.1526
3DFCNN [17]	×2	33.3797	0.9369	4.6173	0.9596	0.0219	3.6197
GDRRN [22]	×2	33.8949	0.9428	4.7006	0.9641	0.0206	3.4179
SSPSR [25]	×2	34.8724	0.9525	4.0143	0.9706	0.0185	3.0734
RFSR [26]	×2	35.5874	0.9586	3.6346	0.9746	0.0171	2.8441
Ours	×2	**35.8185**	**0.9611**	**3.6139**	**0.9758**	**0.0166**	**2.7710**
bicubic	×4	27.3222	0.7151	6.3660	0.8493	0.0451	7.0292
3DFCNN [17]	×4	27.7103	0.7546	6.5670	0.8582	0.0429	6.7438
GDRRN [22]	×4	27.9602	0.7695	7.1670	0.8664	0.0414	6.5732
SSPSR [25]	×4	28.3585	0.7876	6.1930	0.8788	0.0397	6.2665
RFSR [26]	×4	28.6622	0.8010	5.8156	0.8871	0.0383	6.0599
Ours	×4	**28.7923**	**0.8080**	**5.7797**	**0.8904**	**0.0376**	**5.9773**
bicubic	×8	24.3714	0.4531	7.8903	0.6763	0.0646	9.8142
3DFCNN [17]	×8	24.3173	0.4532	8.1556	0.6675	0.0647	9.8779
GDRRN [22]	×8	24.5468	0.4777	8.4873	0.6842	0.0630	9.6256
SSPSR [25]	×8	24.6641	0.4942	8.3048	0.6946	0.0620	9.4980
RFSR [26]	×8	24.8317	0.5037	7.7005	0.7076	0.0610	9.3116
Ours	×8	**24.8461**	**0.5097**	**7.6793**	**0.7086**	**0.0608**	**9.3027**

E. Experimental Results on Pavia Dataset

The Pavia centre dataset is taken by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor during a flight campaign over the center area of Pavia, northern Italy, in 2001. It contains 1096 × 1096 pixels with 102 bands in the spectral range from 430 to 860 nm and the ground sampling distance is 1.3 m. Due to the invalid region, following [25], we crop the informative region with 1096 × 715 × 102 of the original scene. Similar to the setting of the above datasets, we crop the left region of the image into four nonoverlap hyperspectral images with the size of 224 × 224 × 102 for testing. The rest region of the image is cropped to form the overlap patches for training (10% of the training data is randomly selected as a validation set). The spatial size of the training patches and the values of c and o are identical with the setting on the above dataset at scale factors ×2 and ×4. Especially, for the smaller spatial size of Pavia dataset compared to other datasets, we extract the patches with the size of 128 × 128 × 102 by the overlapping of 64 pixels for scale factors ×8 on Pavia dataset.

In Table III, we show the quantitative results of all methods in terms of six evaluation metrics on the Houston test set at scale factors ×2, ×4, and ×8. Due to the limited number of available training samples in the Pavia centre dataset, the performance of each method on this dataset is poor compared to the other datasets. However, in terms of the six evaluation metrics, the proposed MSDformer achieves the better results as well compared with the other state-of-art methods at all scale factors. This demonstrates that our method can be applied to small sample training dataset, which is ideal for hyperspectral remote sensing data. By effectively exploiting the local and global spatial-spectral information, our proposed MSDformer provides the robust similarity in spatial and spectral levels for the hyperspectral images SR.

As shown in Fig. 9, we also visualize the qualitative results of the reconstructed hyperspectral image from Pavia test set at the scale factor ×4. Especially, we treat the 30th, 20th, and 12th bands of the hyperspectral image in the above image as the R-G-B channels for better visualization. The analysis on the above dataset is also suitable for the Pavia dataset. The results of 3DFCNN [17] and GDRRN [22] tend to be blurry. SSPSR [25] and RFSR [26] can produce comparable visual reconstruction results, but there are still some unclear boundaries and severe artifacts. Our proposed MSDformer restores the clearer boundaries and lines than other methods. The advantages of our method can be better observed based on the enlarged view of the details in the red box. Finally, we show the mean spectral difference curve of all test hyperspectral images on Houston dataset at the scale factors ×4 and ×8 in Fig. 10. Our method provides the lowest spectral error while other methods have some spectral distortion.

F. Ablation Study

In this section, we report the effect of different modules on the final SR performance. All variants are trained on the Chikusei dataset at the scale factor ×4. The model parameters and the floating point operations (FLOPs) with the HR image size of 128 × 128 × 128 are calculated on Chikusei dataset at scale factor ×4.

1) Effectiveness of Group Strategy: At the beginning of the network, we divide the spectral bands into subgroups and extract the local spatial-spectral information effectively by following the [25] and [35]. Since the group strategy is thoroughly studied in [25] and [35] and we have analyzed the relevant effects of different settings in the experimental results, we do not study the process of grouping here but rather demonstrate the effectiveness of the grouping strategy. We denote the model without the grouping strategy as Ours w/o GS and train the network in a band-by-band manner. So the complexity of the model will increase with the increase in the number of image spectral bands, which deteriorates

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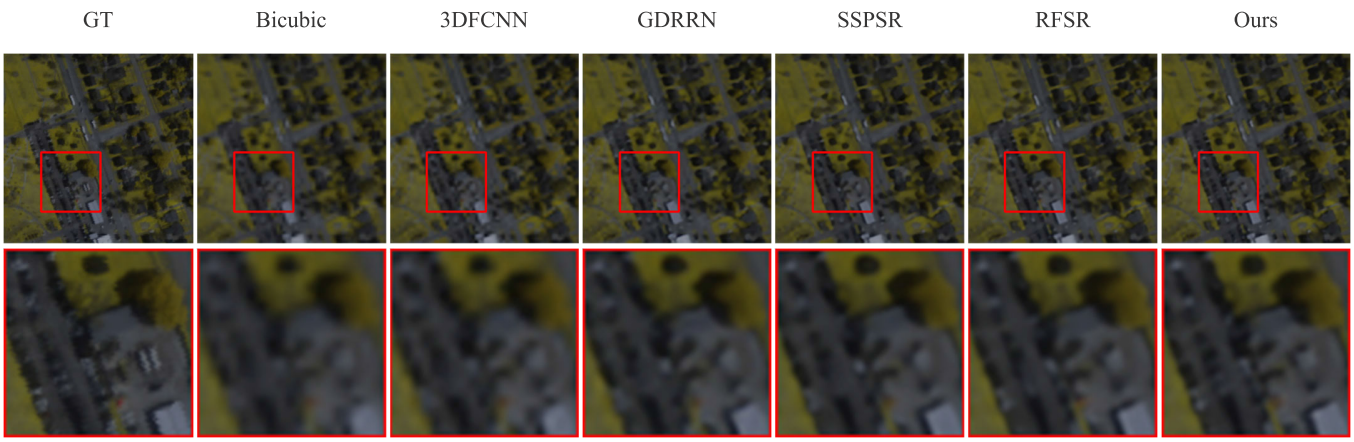


Fig. 7. Reconstructed test hyperspectral image in the Houston dataset with spectral bands 29-26-19 as R-G-B at the scale factor ×4. (From Left to Right) Ground truth, results of bicubic, 3DFCNN [17], GDRRN [22], SSPSR [25], RFSR [26], and the proposed MSDformer method.

TABLE IV

ABLATION STUDY. QUANTITATIVE COMPARISONS OF SOME VARIANTS OF THE PROPOSED METHOD OVER THE TESTING SET ON CHIKUSEI DATASET AT THE SCALE FACTOR ×4

Variant	Params.($\times 10^6$)	FLOPs($\times 10^9$)	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	ERGAS↓
Ours	15.0081	52.3716	**40.0902**	**0.9405**	**2.3981**	**0.9539**	**0.0118**	**5.0818**
Ours w/o GS	28.1286	178.4247	40.0287	0.9397	2.4188	0.9533	0.0118	5.1239
Ours w/o DConv	15.0081	52.3716	39.8901	0.9385	2.4359	0.9518	0.0120	5.2331
Ours w/o CNN	11.5377	26.6738	39.8545	0.9379	2.4681	0.9514	0.0120	5.2570
Ours w/o Transformer	14.1844	51.4977	39.8110	0.9375	2.4811	0.9509	0.0121	5.2937
Ours w/o DFconv	12.7789	50.0890	39.8868	0.9387	2.4635	0.9518	0.0121	5.1991

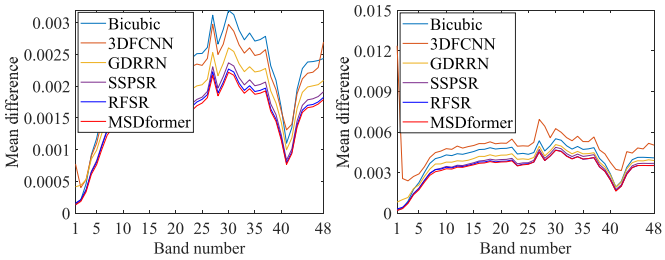


Fig. 8. Mean spectral difference curve of eight test hyperspectral images in the Houston dataset at the scale factors (Left) ×4 and (Right) ×8, respectively.

TABLE V

QUANTITATIVE COMPARISONS OF THE NUMBER OF DCTMS OVER THE TESTING SET ON CHIKUSEI DATASET AT THE SCALE FACTOR ×4. BOLD REPRESENTS THE BEST

Number (N)	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	ERGAS↓
$N=2$	39.9524	0.9385	2.4180	0.9526	0.0120	5.1501
$N=4$	**40.0902**	**0.9405**	**2.3981**	**0.9539**	**0.0118**	**5.0818**
$N=6$	39.9895	0.9395	2.4331	0.9528	0.0119	5.1581
$N=8$	39.9096	0.9387	2.4617	0.9519	0.0120	5.2176

the efficiency of the network. Although the model without grouping strategy preserves some spatial information with comparable PSNR and SSIM, the performance of other metrics deteriorates correspondingly, which demonstrates the effectiveness of grouping strategy. For example, the PSNR of Ours outperforms Ours w/o GS 0.0615 dB and the SAM of Ours is lower than Ours w/o GS by 0.0207.

2) Effectiveness of Multiscale Feature: In MSAM, we employ the three dilated convolutions with different dilation rates to extract the multiscale spatial-spectral features. To verify the effectiveness of the multiscale features, we replace the dilated convolution with the standard convolution to extract the spatial-spectral feature at single scale. The model without dilated convolution is denoted as Ours w/o DConv and its performance is shown in Table IV. All indices decreased considerably at the same time, which proves the importance of local multiscale structure for SHSR. For example, the PSNR of Ours outperforms Ours w/o DConv 0.2001 dB and the SAM of Ours is lower than Ours w/o DConv by 0.0378.

3) Effectiveness of CNN Structure MSAM: In the proposed MSDformer, we extract the local spatial-spectral features by the CNN structure MSAM and further exploit the Transformer module to extract the global spatial-spectral features, which takes advantage of their complementarity. To evaluate the effectiveness of CNN structure, we design a network consisting entirely of the Transformer module DCTM. We denote the above Transformer network as Ours w/o CNN in Table IV. Although the spectral similarity is slightly preserved by the Transformer module, the spatial similarity has deteriorated severely, which demonstrates the significance of the designed MSAM. For example, the PSNR of Ours outperforms Ours w/o CNN 0.2357 dB and the SAM of Ours is lower than Ours w/o CNN by 0.07.

4) Effectiveness of Transformer Module DCTM: In the proposed MSDformer, we extract the global spatial-spectral features by the Transformer module DCTM and further

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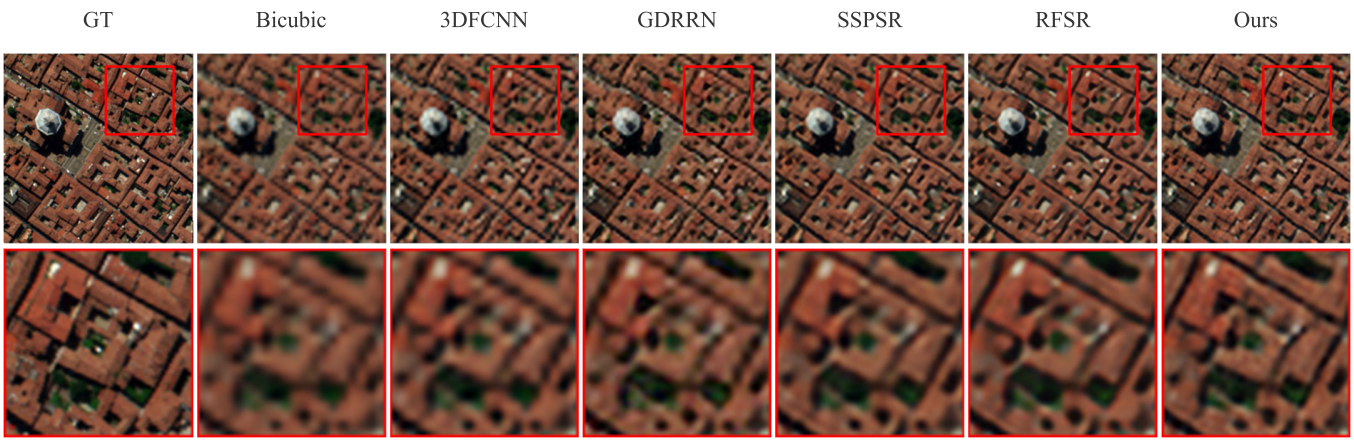


Fig. 9. Reconstructed test hyperspectral image in the Pavia dataset with spectral bands 30-20-12 as R-G-B at the scale factor ×4. (From Left to Right) Ground truth, results of bicubic, 3DFCNN [17], GDRRN [22], SSPSR [25], RFSR [26], and the proposed MSDformer method.

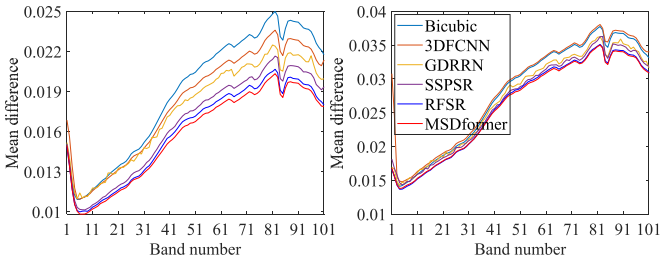


Fig. 10. Mean spectral difference curve of four test hyperspectral images in the Pavia dataset at the scale factors (Left) ×4 and (Right) ×8, respectively.

TABLE VI

QUANTITATIVE PERFORMANCE (AVERAGE PSNR/SSIM/SAM) OF DIFFERENT LOSS FUNCTIONS EVALUATED OVER FOUR TESTING IMAGES OF CHIKUSEI DATASET AT THE SCALE FACTOR ×4

$\mathcal{L}_1$	$\mathcal{L}_{spe}$	$\mathcal{L}_{gra}$	PSNR↑	SSIM↑	SAM↓
✓			40.0502	0.9402	2.4861
✓	✓		39.7668	0.9364	2.4528
✓		✓	39.9421	0.9387	2.5305
✓	✓	✓	40.0902	0.9405	2.3981

exploit the Transformer module to explore the long-range spectral dependencies from the local features of the previous stage, which improves the representation ability of the model. To evaluate the effectiveness of the Transformer module, we design a network consisting entirely of the CNN module. Specifically, we replace the Transformer network with the CA module which is commonly used in SR and denote it as Ours w/o Transformer in Table IV. The values of all metrics have deteriorated significantly, which demonstrates the significance of the designed DCTM. For example, the PSNR of Ours outperforms Ours w/o Transformer 0.2792 dB and the SAM of Ours is lower than Ours w/o Transformer by 0.083.

5) Effectiveness of Deformable Convolution: For exploring the various diverse long-range correlations, we design a DMSA to learn the specific weights for different spectral characteristics, which fully exploits the special spectral patterns of hyperspectral images. To evaluate the effectiveness of deformable convolution in DMSA, we design a network with the standard MHSA which is denoted as Ours w/o DFconv in Table IV. We can find that the spatial and structural similarity generated by this variant is seriously poor. For example, the PSNR of Ours outperforms Ours w/o DFconv 0.2034 dB and the SAM of Ours is lower than Ours w/o DFconv by 0.0654. It demonstrate that the various global spatial-spectral dependencies is beneficial for the hyperspectral image SR.

6) Analysis of the Number of DCTMs: To extract the global spatial-spectral information, we adopt N DCTMs in the network. We study the influence of the number of recursive DCTMs on the hyperspectral SR performance. The experimental results are shown in Table V. When using fewer modules ($N = 2$), all quantitative indices are the worst. When the number of DCTMs is further set to 4 ($N = 4$), the quantitative indices such as PSNR and SSIM have significantly increased. And when we further increase the number of recursive modules, the SR performance has decreased instead, probably because the deeper model requires more training data.

7) Hyperparameter Analysis: We investigate the effectiveness of different combinations of losses on the experimental results and three representative quantitative metrics are shown in Table VI. When only using l_1 loss, our method can achieve the second best spatialwise SR results (i.e., PSNR and SSIM) as well as poor spectral performance (i.e., SAM). When the single spectral loss is added, the spectral similarity increases accordingly at the expense of spatial accuracy. When the single gradient loss is added to enhance local sharpening, the structural similarity increases compared to the above case of adding spectral loss. Furthermore, if using three losses simultaneously, our network will provide the best results at both spatial and spectral levels. By using the mentioned loss strategy in this article, our method preserves excellent spatial-spectral similarity. We also study the effect of combinations of different loss weight for the results. In the previous methods [26], λ_1 is set as 0.5 and λ_2 is set as 0.1. So we compare the PSNR values of different combinations of loss weights on Chikusei based on the above values. As shown in Fig. 11, we can find that the PSNR value achieves the maximum when $\lambda_1 = 0.5$ and $\lambda_2 = 0.1$, which makes us choose the settings.

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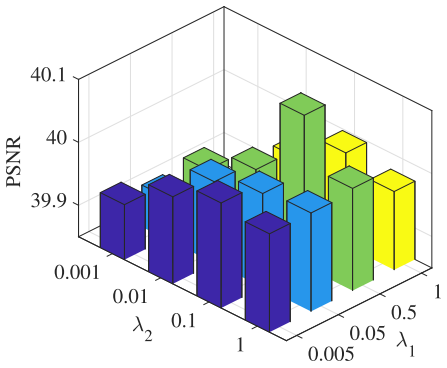


Fig. 11. PSNR of different combinations of loss weights on Chikusei dataset.

V. Conclusion

In this article, MSDformer is proposed for hyperspectral image SR to learn the local and global spatial-spectral relevance and discrepancy, which takes advantage of the CNN structure and Transformer architecture. In MSDformer, we first design an MSAM to extract the local multiscale spatial-spectral feature in a group-by-group manner, which employs the spatial redundancy and spectral difference between spectral groups. Specifically, the grouping strategy is proven to be effective in improving results with slightly increased parameters and computational costs in this network. Furthermore, we design a Transformer-based module DCTM by adopting the deformable convolution in the MHSA to capture the diverse long-range spectral dependencies from the extracted local feature. Finally, we conduct the extensive ablation experiments to verify the effectiveness of the module in MSDformer. The qualitative and quantitative results on common hyperspectral image datasets have demonstrated the superiority of our method over the state-of-the-art methods at different scale factors.

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